Chapter 16 Review — Vector Calculus

MTH 310, Spring 2026

1. The Big Picture: Four Theorems, One Pattern

Every major theorem in vector calculus relates a **boundary integral** to an **interior integral** of a derivative:

Theorem	Boundary	Interior	Derivative
FTC (Calc I)	two endpoints	interval $[a, b]$	f'
Green's Thm (16.4)	curve C in plane	flat region D	$Q_x - P_y$
Stokes' Thm (16.8)	space curve C	surface S in 3D	$\operatorname{curl} \vec{F}$
Divergence Thm (16.9)	closed surface S	solid E	$\operatorname{div} \vec{F}$

2. Vector Fields and Line Integrals

Scalar line integral (16.2)

$$\int_C f\,ds = \int_a^b f(\vec{r}(t))\,|\vec{r}\,'(t)|\,dt.$$

Use for: total mass of a wire, average value along a curve.

Vector line integral / work (16.2)

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t)\,dt.$$

Use for: work, circulation along C.

Recipe (both kinds): (1) parametrize C, (2) compute $\vec{r}\,'(t)$, (3) plug in, (4) integrate.

3. Conservative Fields and FTLI (16.3, 16.5)

$\vec{F}$ is **conservative** on a simply connected domain $\iff$ $\vec{F} = \nabla f$ for some potential f $\iff$ $\operatorname{curl} \vec{F} = \vec{0}$ (**2D test:** $P_y = Q_x$). When conservative,

$$\int_C \vec{F} \cdot d\vec{r} = f(\text{end}) - f(\text{start}) \quad \text{(FTLI)}.$$

Finding a potential. Solve $f_x = P$, then $f_y = Q$, then (in 3D) $f_z = R$, picking up integration "constants" that depend on the remaining variables.

4. Green's Theorem (16.4)

For a simple closed CCW curve C bounding a region D in the plane:

$$\oint_C P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dA.$$

When to use: the line integral is hard but $Q_x - P_y$ is simple; or one of P, Q has a non-elementary antiderivative (like e^{x^2} or $\sin(y^2)$) that drops out when you differentiate.

Area formula: $\text{Area}(D) = \dfrac{1}{2}\oint_C (x\,dy - y\,dx)$.

5. Two Ways to Handle a Surface — the Big Comparison

This is the formula bottleneck of the chapter. Every surface problem reduces to choosing one of these two setups:

	Graph form $z = g(x, y)$	**Parametric form** $\vec{r}(u, v)$
Parametrization	$\vec{r}(x, y) = \langle x, y, g(x, y)\rangle$	$\vec{r}(u, v) = \langle x(u, v), y(u, v), z(u, v)\rangle$
Tangent vectors	$\vec{r}_x = \langle 1, 0, g_x\rangle$, $\vec{r}_y = \langle 0, 1, g_y\rangle$	$\vec{r}_u, \vec{r}_v$ (compute by differentiating)
Cross product	$\vec{r}_x \times \vec{r}_y = \langle -g_x, -g_y, 1\rangle$ (upward)	$\vec{r}_u \times \vec{r}_v$ (compute)
Area element dS	$\sqrt{g_x^2 + g_y^2 + 1}\, dA$	$\lvert\vec{r}_u \times \vec{r}_v\rvert\, du\, dv$
Vector area $\vec{n}\, dS$	$\langle -g_x, -g_y, 1\rangle\, dA$	$(\vec{r}_u \times \vec{r}_v)\, du\, dv$
Domain of integration	projection onto xy-plane	rectangle/region in uv-plane

When to use which?

- **Graph** $z = g(x, y)$: when the surface is given as "$z =$ something". Easiest for paraboloids, planes, and graphs over a rectangle/disk in xy.
- **Parametric** $\vec{r}(u, v)$: when the surface is given parametrically, or when the surface is a sphere/cylinder/cone where graph form is awkward.

6. Spherical Parameterization of Spheres

For a sphere of radius R centered at the origin, the spherical parameterization is so important it deserves its own section:

$$\vec{r}(\theta, \phi) = \langle R\sin\phi\cos\theta,\ R\sin\phi\sin\theta,\ R\cos\phi\rangle, \quad 0 \le \theta \le 2\pi,\ 0 \le \phi \le \pi.$$

Easy-to-remember facts:

- **Outward unit normal** $\widehat{n} = \dfrac{\langle x, y, z\rangle}{R}$ (just the position vector, normalized).
- **Cross product magnitude:** $\lvert\vec{r}_\phi \times \vec{r}_\theta\rvert = R^2 \sin\phi$.
- **Cross product itself:** $\vec{r}_\phi \times \vec{r}_\theta = R^2 \sin\phi \cdot \widehat{n}$.

Cylindrical cutoffs. If you're integrating over the part of the sphere inside the cylinder $x^2 + y^2 = a^2$ (with $a \le R$), the constraint becomes $R\sin\phi \le a$, i.e. $\sin\phi \le a/R$, so ϕ runs 0 to $\arcsin(a/R)$.

Plane cutoffs. For "above $z = h$" on a sphere of radius R: $R\cos\phi \ge h$, so ϕ runs 0 to $\arccos(h/R)$.

7. Surface Integrals (16.6, 16.7)

Scalar surface integral (mass on a curved sheet, total of f over S):

Graph $z = g(x, y)$:

$$\iint_S f\, dS = \iint_D f(x, y, g)\sqrt{g_x^2 + g_y^2 + 1}\, dA.$$

Parametric $\vec{r}(u, v)$:

$$\iint_S f\, dS = \iint_D f(\vec{r})\, \lvert\vec{r}_u \times \vec{r}_v\rvert\, du\, dv.$$

Flux integral (flow of $\vec{F}$ through the surface in the direction of $\vec{n}$):

Graph $z = g(x, y)$, upward:

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D (-Pg_x - Qg_y + R)\, dA.$$

(For downward, flip the sign.)

Parametric $\vec{r}(u, v)$:

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D \vec{F}(\vec{r}) \cdot (\vec{r}_u \times \vec{r}_v)\, du\, dv.$$

(For opposite orientation, flip the sign or swap the cross-product order.)

Note on flux for graphs: no square root! The $|\times|$ from dS cancels the denominator of $\widehat{\vec{n}}$.

8. Stokes' Theorem (16.8)

For a smooth oriented surface S with boundary curve C (right-hand rule):

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \operatorname{curl} \vec{F} \cdot d\vec{S}.$$

When to use: you have a line integral over a 3D closed curve and direct parametrization is hard — choose any convenient surface S with that boundary, and compute the curl flux instead. You may also use Stokes' *in reverse*: a complicated $\iint_S \operatorname{curl} \vec{F} \cdot d\vec{S}$ becomes a line integral on the boundary.

9. Divergence Theorem (16.9)

For a closed surface S bounding a solid E (outward orientation):

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_E \operatorname{div} \vec{F}\, dV.$$

When to use: flux through a *closed* surface is a giveaway. Especially powerful when $\operatorname{div} \vec{F}$ is simple (constant, polynomial) and the solid has nice cylindrical/spherical symmetry.

10. Quick Reference: Which Theorem When?

If you see...	**Try...**
$\int_C \nabla f \cdot d\vec{r}$	FTLI: $f(\text{end}) - f(\text{start})$
Closed curve in 2D, line integral	Green's Theorem (especially if the integrand has nasty e^{x^2}, $\sin(y^2)$ terms)
Closed curve in 3D, line integral	Stokes' Theorem
Closed surface, flux integral	Divergence Theorem
Open surface, flux integral	Stokes' (in reverse), or parametrize directly

Common simplifications worth scanning for before grinding:

- Symmetry: odd functions over symmetric regions vanish (a $2x$ over a disk centered at origin is 0).
- Constant divergence or curl: $\iiint c\, dV = c \cdot \text{Vol}$, $\iint c\, dA = c \cdot \text{Area}$.
- Conservative? Closed loop $\Rightarrow$ integral is 0.
- Can you replace a complicated surface by a simpler one with the same boundary? (For Stokes', yes.)